

# Recommendation **ITU-T K.100 (08/2024)**

SERIES K: Protection against interference

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**Measurement of radio frequency  
electromagnetic fields to determine compliance  
with human exposure limits when a base  
station is put into operation**



# Recommendation ITU-T K.100

## Measurement of radio frequency electromagnetic fields to determine compliance with human exposure limits when a base station is put into operation

### Summary

Recommendation ITU-T K.100 provides information on measurement techniques and procedures for assessing compliance with the general public electromagnetic field (EMF) exposure limits when a new base station (BS) is put into operation, taking into account effects of the environment and other relevant radio frequency sources present in its surroundings.

### History\*

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### Keywords

Actual maximum approach, base station, compliance, electromagnetic fields, exposure assessment, put into operation.

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## **Recommendation ITU-T K.100**

### **Measurement of radio frequency electromagnetic fields to determine compliance with human exposure limits when a base station is put into operation**

#### **1 Scope**

This Recommendation specifies the measurement procedure to assess compliance with general public electromagnetic field (EMF) exposure limits for a base station (BS) operating in the frequency range 100 MHz-100 GHz when it is put into operation in its operational environment.

Simplified assessment procedures are provided to identify those installations that are known to be compliant with EMF exposure limits without measurements.

With its specific focus on measurements, this Recommendation complements the existing ITU-T K-series Recommendations.

Contact currents due to contact with conductive objects irradiated by EMFs, lie outside the scope of this Recommendation. For commercial market BS products, there could be other requirements specified by the manufacturer that might need to be fulfilled. For such types of product testing, this Recommendation also does not apply.

The entity conducting a radiofrequency (RF) exposure assessment needs to be aware of the applicable national laws, standards or guidelines on exposure limits to EMF and assessment procedures.

#### **2 References**

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

- [ITU-T K.52] Recommendation ITU-T K.52 (2024), *Guidance on complying with limits for human exposure to electromagnetic fields*.
- [ITU-T K.61] Recommendation ITU-T K.61 (2018), *Guidance on measurement and numerical prediction of electromagnetic fields for compliance with human exposure limits for telecommunication installations*.
- [ITU-T K.70] Recommendation ITU-T K.70 (2020), *Mitigation techniques to limit human exposure to emfs in the vicinity of radiocommunication stations*.
- [ITU-T K.91] Recommendation ITU-T K.91 (2024), *Guidance for assessment, evaluation and monitoring of human exposure to radio frequency electromagnetic fields*.
- [IEC 62232] IEC 62232:2022, *Determination of RF field strength, power density and SAR in the vicinity of base stations for the purpose of evaluating human exposure*.
- [IEC 62479] IEC 62479:2010, *Assessment of the compliance of low-power electronic and electrical equipment with the basic restrictions related to human exposure to electromagnetic fields (10 MHz to 300 GHz)*.

## 3 Definitions

### 3.1 Terms defined elsewhere

This Recommendation uses the following terms defined elsewhere:

**3.1.1 actual value** [IEC 62232]: Time-averaged value of transmitted power or equivalent isotropically radiated power (EIRP) of the equipment under test (EUT) using the averaging time specified in the applicable radio frequency (RF) exposure limits, e.g., 6 minutes using [b-ICNIRP].

**3.1.2 actual maximum (max) value** [IEC 62232]: Maximum value reached by the actual value when recorded by the EUT during a given period (e.g., week or month).

**3.1.3 actual maximum (max) approach** [IEC 62232]: Evaluation method of RF exposure taking into account the actual maximum thereof.

**3.1.4 actual maximum (max) threshold** [IEC 62232]: Upper limiting value during base station (BS) operation of the actual transmitted power or EIRP configured by the EUT, for example using transmitted power or EIRP control features and tools implemented in the EUT.

**3.1.5 ambient source** [IEC 62232]: A radio frequency (RF) source operating in the frequency range from 8.3 kHz to 300 GHz generating electromagnetic fields other than the emission from the equipment under test (EUT).

**3.1.6 assessment domain boundary (ADB)** [IEC 62232]: Surface surrounding an antenna of the equipment under test (EUT) outside of which the TER is lower than 0.05 and measurements do not need to be conducted. The ADB defines the maximum possible measurement area where the source is considered to be relevant.

**3.1.7 average (temporal) power ( $P_{avg}$ )**: Time-averaged rate of energy transfer defined by:

$$P_{avg} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} P(t) dt$$

where  $t_1$  and  $t_2$  are the start and stop time of the exposure. The period  $t_1 - t_2$  is the exposure duration time.

**3.1.8 averaging time ( $T_{avg}$ )** [IEC 62232]: Time duration specified in the applicable exposure limit guidelines over which exposure is averaged.

**3.1.9 base station (BS)** [IEC 62232]: Equipment, permanently or temporarily installed, including the radio transmitter(s) and associated antenna(s) as used in wireless telecommunications networks.

**3.1.10 compliance boundary (CB)** [IEC 62232]: Surface defining a volume outside which the radio frequency (RF) exposure from the equipment under test (EUT) is below the applicable exposure limit.

**3.1.11 compliance distance (CD)** [IEC 62232]: Distance from the EUT antenna to the compliance boundary for a stated direction and set of transmission conditions.

NOTE 1 – In the absence of a stated direction, the compliance distance is the maximum distance from the EUT antenna to the compliance boundary.

NOTE 2 – The abbreviation ( $D$ ) used in some documents for the compliance distance. However, this can be confused with "Directivity". This is why CD is preferred.

**3.1.12 configured envelope radiation pattern** [IEC 62232]: Non-physical radiation pattern obtained by taking, for each azimuth and elevation direction, the maximum performance among the set of the radiation patterns of the base station antenna configured during operation.

**3.1.13 configured maximum (max) value** [IEC 62232]: RMS maximum of the value (e.g., transmitted power or EIRP) as configured in the network manager of the base station.



**3.1.14 domain of investigation (DI)** [IEC 62232]: Sub-domain within the assessment domain boundary (ADB) to which the general public have access.

**3.1.15 equivalent isotropically radiated power (EIRP)** [IEC 62232]: The product of the input power into an antenna and the antenna gain in a given direction relative to an isotropic antenna.

**3.1.16 exposure ratio (ER)** [IEC 62232]: The ratio of the exposure metric and the applicable exposure limit, both expressed in terms of power.

NOTE – For assessment against reference levels:

$$ER = \max [(E/E_{\text{lim}})^2, (H/H_{\text{lim}})^2]$$

In the far-field:

$$ER = (E/E_{\text{lim}})^2 = (H/H_{\text{lim}})^2 = S/S_{\text{lim}}$$

where  $S$ ,  $E$ , and  $H$  are the root mean square (r.m.s.) values, respectively, of power density, electric field strength and magnetic field strength measured at frequency  $f$ .  $S_{\text{lim}}$ ,  $E_{\text{lim}}$  and  $H_{\text{lim}}$  are the corresponding limits at the same frequency.

When exposure is evaluated for a certain frequency band (the total power density or the field strength within the frequency interval  $[f_{\text{min}}, f_{\text{max}}]$  is assessed),  $S_{\text{lim}}$ ,  $E_{\text{lim}}$  or  $H_{\text{lim}}$  are chosen as the most stringent limits within the band.

**3.1.17 multiple-input, multiple-output (MIMO)** [IEC 62232]: Technique used to increase the communication throughput by using transmission of two or more different data streams on two or more different antennas, using the same resources in both frequency and time, separated only by the use of different reference signals, and to be received by two or more antennas.

**3.1.18 point of maximum exposure (POME)** [IEC 62232]: location of the maximum RF exposure when performing a slow scan over a measurement area at a height of 1,5 m above the walking or standing surface.

**3.1.19 power reduction factor ( $F_{\text{PR}}$ )** [IEC 62232]: Factor applied to the product of the configured maximum transmitted power or EIRP and the technology duty-cycle factor in order to obtain the actual maximum transmitted power or EIRP, or the actual maximum power or EIRP threshold, when a control feature is implemented.

NOTE – When applied to EIRP, the power reduction factor includes variations in both the transmitted power and the antenna gain.

**3.1.20 relevant source** [IEC 62232]: A radio source, in the frequency range 8.3 kHz to 300 GHz, with an exposure ratio (ER) equal or greater than 0.05 to be considered when performing an RF exposure assessment.

NOTE – It is possible that applicable regulations consider ER criteria lower than 0.05 to be relevant (e.g., 0.01 of the measured TER).

**3.1.21 segment (of a BS sector or a BS cell)** [IEC 62232]: Solid angle in front of a BS sector or a BS cell described by an azimuth aperture range (delimited by an initial angle  $\phi_i$  and a final angle  $\phi_{i+1}$ ) and elevation aperture range (delimited by an initial angle  $\theta_j$  and a final angle  $\theta_{j+1}$ )

**3.1.22 technology duty-cycle factor ( $F_{\text{TDC}}$ )** [IEC 62232]: Ratio of the time that a transmitter transmits over a specified period.

**3.1.23 total exposure ratio (TER):** The sum of exposure ratios (ERs) of the equipment under test (EUT) and other relevant sources.

## 3.2 Terms defined in this Recommendation

This Recommendation defines the following terms:

**3.2.1 antenna gain (G):** Ratio of power radiated per unit solid angle multiplied by  $4\pi$  to the total input power. Gain is frequently expressed in decibels with respect to an isotropic antenna (dBi). The equation defining gain is:

$$G(\theta, \varphi) = \frac{4\pi}{P_{in}} \frac{dP_r}{d\Omega}$$

where:

$\theta, \varphi$ : the angles in a polar coordinate system

$P_r$ : the radiated power along the  $(\theta, \varphi)$  direction

$P_{in}$ : the total input power

$\Omega$ : elementary solid angle along the direction of observation.

**3.2.2 equipment under test (EUT):** For the purpose of this Recommendation, the EUT is the base station that is put into operation, including all transmitting antennas (operating in the frequency range 100 MHz to 300 GHz).

**3.2.3 far-field formula:** Formula that can be used in the far-field to evaluate the power density,  $S$ :

$$S = \frac{PG_{\theta, \varphi}}{4\pi d^2}$$

where  $P$  is the transmitted power,  $G_{\theta, \varphi}$  is the gain of the antenna in the direction  $(\theta, \varphi)$  and  $d$  is the distance from the antenna to the evaluation point. The associated electric field strength,  $E$ , and magnetic field strength,  $H$ , can be evaluated as follows:

$$E = \frac{\sqrt{30PG_{\theta, \varphi}}}{d},$$
$$H = \frac{E}{\eta_0},$$

where  $\eta_0 \approx 377 \Omega$ .

If the power density is evaluated in the direction of maximum antenna gain:

$$S = \frac{EIRP}{4\pi d^2}.$$

**3.2.4 isotropic antenna:** A hypothetical, lossless antenna having equal radiation intensity in all directions.

**3.2.5 main lobe:** The radiation lobe containing the direction of maximum radiation. In certain antennas, such as multilobed or split-beam antennas, there may be more than one major lobe.

**3.2.6 reactive near-field region:** The reactive near field of an antenna with maximum extension  $D$  is in this context defined as  $\max(\lambda, D, D^2/4\lambda)$ , where  $\lambda$  denotes the free space wavelength.

**3.2.7 side lobe:** A radiation lobe in any direction other than the main lobe.

## 4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

ADB                      Assessment Domain Boundary

|          |                                            |
|----------|--------------------------------------------|
| BCCH     | Broadcast Control Channel                  |
| BS       | Base Station                               |
| CB       | Compliance Boundary                        |
| CDF      | Cumulative Distribution Function           |
| CDMA     | Code Division Multiple Access              |
| CPICH    | Common Pilot Channel                       |
| DAB      | Digital Audio Broadcasting                 |
| DI       | Domain of Investigation                    |
| DVB-T    | Digital Video Broadcasting-Terrestrial     |
| EIRP     | Equivalent Isotropically Radiated Power    |
| EMF      | Electromagnetic Field                      |
| ER       | Exposure Ratio                             |
| EUT      | Equipment Under Test                       |
| FDMA     | Frequency Division Multiple Access         |
| FM       | Frequency Modulation                       |
| GSM      | Global System for Mobile Communications    |
| LOS      | Line Of Sight                              |
| LTE      | Long-Term Evolution                        |
| MCCH     | Multicast Control Channel                  |
| MIMO     | Multiple-Input Multiple-Output             |
| NR       | New Radio                                  |
| OFDM     | Orthogonal Frequency-Division Multiplexing |
| PBCH     | Physical Broadcast Channel                 |
| PSS      | Primary Synchronization Signal             |
| RBW      | Resolution Bandwidth                       |
| RF       | Radio Frequency                            |
| r.m.s.   | root mean square                           |
| SAR      | Specific Absorption Rate                   |
| SD       | Standard Deviation                         |
| SSB      | Synchronisation Signal Block               |
| SSS      | Secondary Synchronization Signal           |
| TACS     | Total Access Communications System         |
| TDMA     | Time Division Multiple Access              |
| TER      | Total Exposure Ratio                       |
| TETRA    | Terrestrial Trunked Radio                  |
| TETRAPOL | Terrestrial Trunked Radio for Police       |
| UMTS     | Universal Mobile Telecommunication System  |

|       |                                        |
|-------|----------------------------------------|
| WCDMA | Wideband Code Division Multiple Access |
| Wi-Fi | Wireless Fidelity                      |

## **5 Conventions**

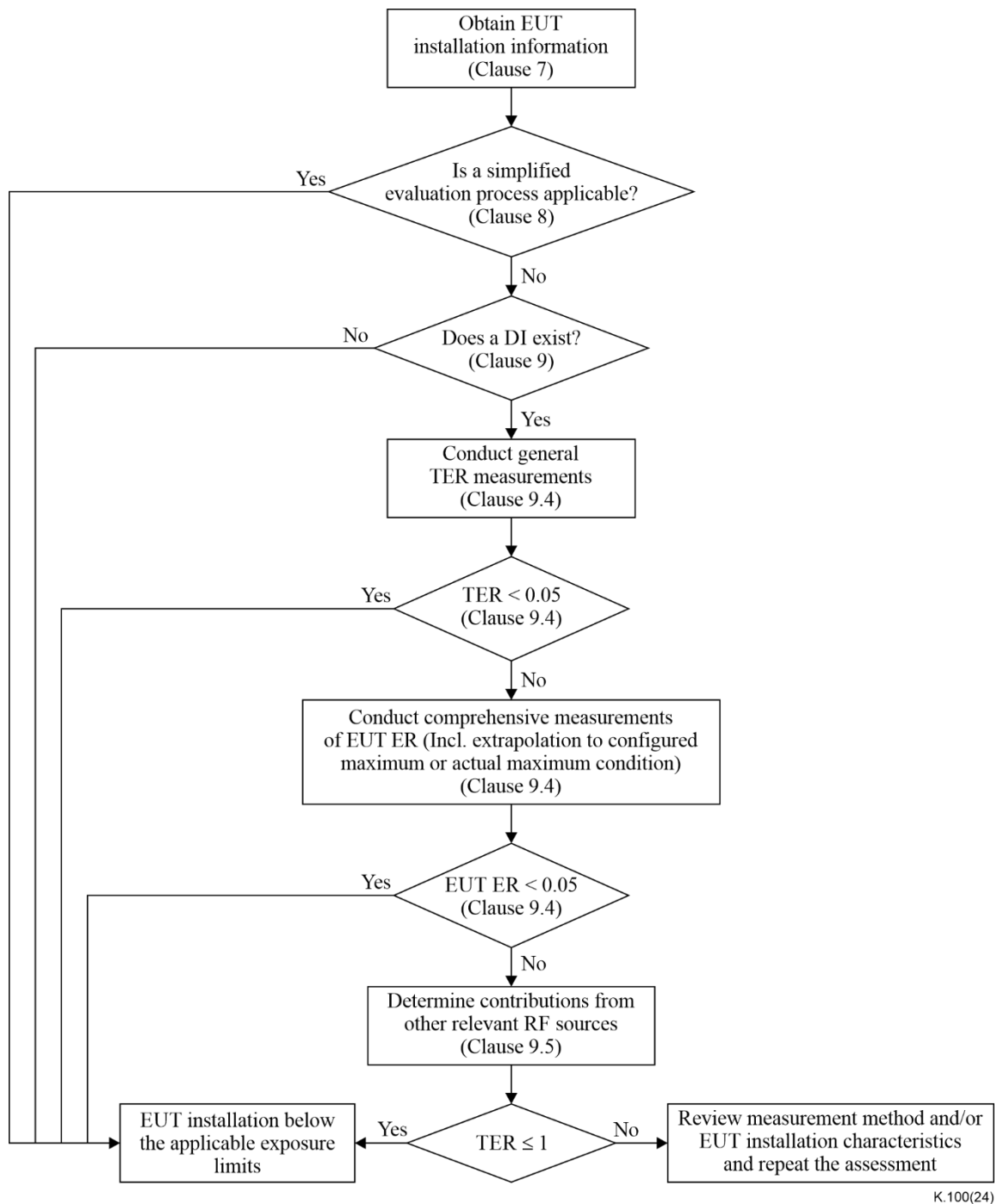
None.

## **6 Assessment procedure**

### **6.1 General process**

When a base station (BS), is installed and put into operation at a site, the procedure described in Figure 6-1 shall be used to assess compliance with exposure limits. In order to allow for accurate and efficient assessments, different routes are possible depending on the characteristic of the equipment under test (EUT) or on the installation type. The compliance assessment process can be implemented using the actual maximum equivalent isotropically radiated power (EIRP) threshold of the BS site sector, cell, or segment and following the guidelines described in [IEC 62232]. The actual maximum EIRP threshold can be derived from the actual maximum transmitted power using the maximum gain of the antenna or individual antenna beams for massive multiple-input multiple-output (MIMO).

In some specific cases, compliance with relevant exposure limits can be assessed without the necessity to conduct measurements (e.g., because of the low power transmitted, or because of the position or orientation of the transmitters or antennas with respect to areas accessible to the general public, or because simpler calculation methods can be used [ITU-T K.52]). Both broadband and frequency-selective equipment can be used for the assessment (see clause 9.2). Measurements conducted with broadband equipment, however, might lead to overly conservative results. If the exposure level in areas accessible to the general public is found to be above the limits by means of broadband measurements, then compliance should be verified with frequency-selective equipment. Otherwise, the mitigation techniques described in [ITU-T K.70] shall be applied. The step-by-step procedure of Figure 6-1 is specified in clauses 7 to 10.



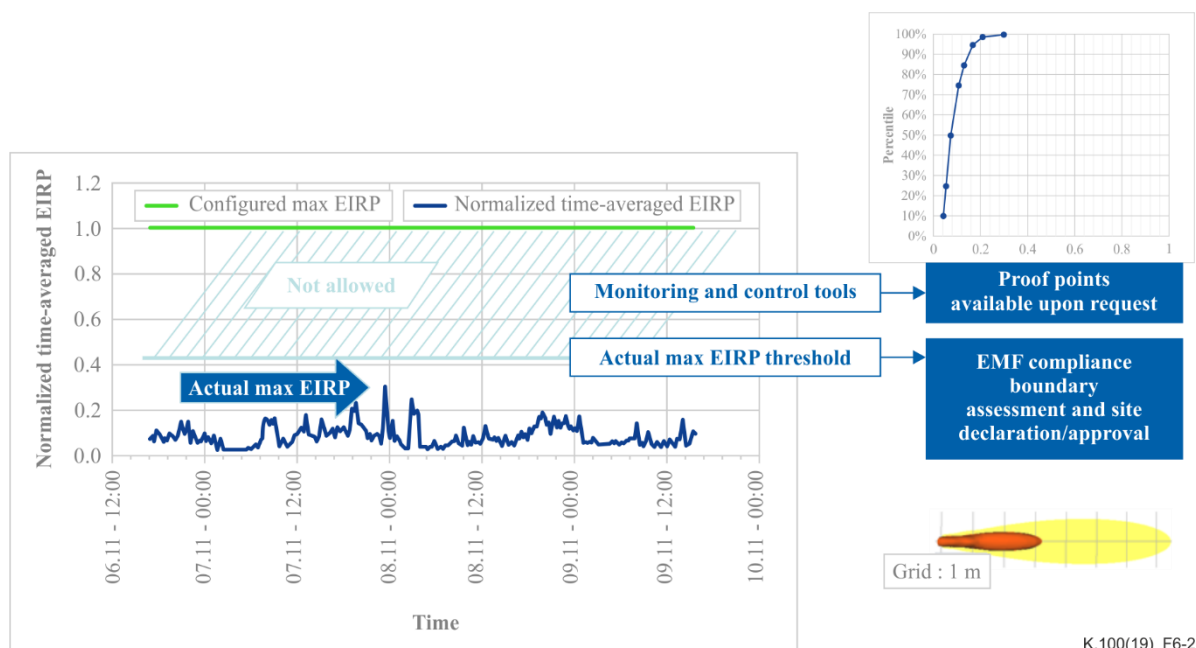
**Figure 6-1 – Flowchart describing the measurement procedure specified in this Recommendation (derived from [IEC 62232])**

## 6.2 Method based on the actual maximum transmitted power or EIRP

The method based on actual maximum transmitted power or EIRP provides for more accurate assessment of the real maximum exposure from a base station using the parameters of the site. It avoids the unrealistic over-estimation of other approaches. The rationale for using the actual maximum approach is described in [IEC 62232] and [b-IEC TR 62669].

Figure 6-2 shows an example of BS site sector based on the actual max EIRP. The RF exposure compliance boundary is defined based on the actual maximum EIRP threshold configured by the BS. The BS is implementing monitoring and/or control features (MCF) ensuring that the actual EIRP does not exceed this actual maximum EIRP threshold during operation. The recording of the actual EIRP

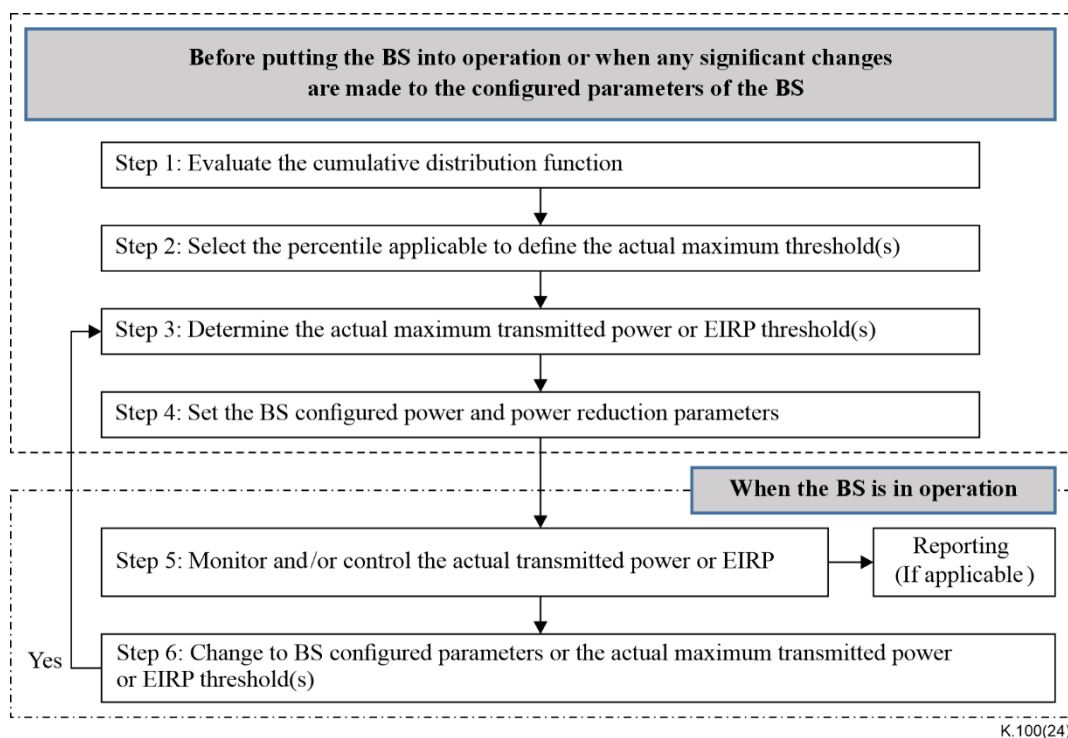
over time is summarized in a cumulative distribution function (CDF) that provide the actual max EIRP (100th percentile) and other lower percentiles (95th or 99th)



**Figure 6-2 – Example of implementation of compliance based on actual max EIRP**

The process described in Figure 6-3 should be used to assess BS installation compliance based on the actual maximum approach.

NOTE – Step 1 and Step 2 can be optional. For Step 4, it is possible that the network operator sets the BS configured power and, if applicable, the power reduction parameters per sector, cell or segment such that the actual transmitted power or EIRP does not exceed the actual maximum transmitted power or EIRP threshold(s) specified in Step 3 during operation.



**Figure 6-3 – Flow chart for product installation compliance based on the actual maximum transmitted power or EIRP threshold(s) (from [IEC 62232])**

[b-IEC TR 62669] provides measurement and simulation case studies for the implementation of the actual maximum approach, including validation of monitoring and control features.

## 7 Equipment under test data collection

The EUT information required for performing the EUT compliance assessment is described in [IEC 62232]. This includes the following

- technology (including the duty cycle factor), frequency band and channel bandwidth<sup>1</sup>
- transmitted power and antenna gain, or actual maximum EIRP threshold (directly or derived from the maximum configured transmitted power) of the BS site sector, cell, or segment for each EUT antenna.

When *comprehensive* assessments supporting extrapolation to maximum or actual maximum traffic are needed (see clause 9.5), the following information may be used for the global system for mobile communications (GSM), wideband code division multiple access (WCDMA) and long-term evolution (LTE):

- GSM: Central frequency of the broadcast control channel (BCCH) and number of carriers (channels) used by the EUT.
- WCDMA: Common pilot channel (CPICH) frequency and power level relative to total power.
- LTE: Centre frequency of the EUT channels and bandwidth where the used channels include the CRS channel(s) and the boosting factor for the CRS.

<sup>1</sup> For multicarrier and multiple radio access technology, BS information shall be gathered for every technology and frequency band to put into operation.

- NR: centre frequency of the EUT used channels and bandwidth, where the used channels include the synchronisation signal block (SSB) (i.e., PSS, SSS, PBCH and PBCH-DMRS) and CSI-RS channel(s) and the boosting factor for the SSB.

## 8 Simplified assessment procedures

Simplified assessment procedures [IEC 62232] are provided to identify an EUT that is known to be compliant with relevant exposure limits without the necessity to follow general or comprehensive exposure assessment processes. This is relevant, for example, because of the low power transmitted or because of the position of the transmitters or antennas (EUT and relevant sources) with respect to the general public.

EUT installation evaluation is not required if the EUT transmitted power complies with [IEC 62479], or if the EUT compliance distance(s) are zero, see installation class E0 in Table 8-1.

The simplified assessment procedure is based on the knowledge of the EUT equivalent isotropically radiated power (EIRP), and depending on the EIRP level, antenna installation characteristics, such as mounting height, main lobe direction and distance to other ambient sources, If the EUT installation characteristics satisfy the installation classes described in Table 8-1<sup>2</sup>, which are derived from [IEC 62232], the EUT installation is deemed to be compliant. Otherwise, an RF EMF exposure assessment shall be performed using the procedures described in clauses 9 to 10<sup>3</sup>.

**Table 8-1 – Example of product installation classes where a simplified evaluation process is applicable (based on [b-ICNIRP] general public limits)**

| Instal. class | EIRP <sup>a</sup> (W) | EIRP <sup>a</sup> (dBm) | Installation criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------|-----------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E0            | N/A                   | N/A                     | The product complies with [IEC 62479] or the EUT compliance distance(s) are zero. No specific requirement for EUT installation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| E2            | $\leq 2$              | $\leq 33$               | The product is installed in accordance with instructions from the manufacturer and/or entity putting into operation. Compliance with the exposure limits is generally obtained at zero distance or within a few centimetres.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| E10           | $\leq 10$             | $\leq 40$               | The EUT is installed in accordance with instructions from the manufacturer and/or entity putting into operation and the lowest radiating part of the antenna(s) is at a minimum height of 2.2 m above the walking or standing surface accessible by the general public.                                                                                                                                                                                                                                                                                                                                                                                        |
| E100          | $\leq 100$            | $\leq 50$               | The EUT is installed in accordance with instructions from the manufacturer and/or entity putting into operation and: <ul style="list-style-type: none"> <li>(i) the lowest radiating part of the antenna(s) is at a minimum height of 2.5 m above the walking or standing surface accessible by the general public,</li> <li>(ii) the minimum distance to areas accessible to the general public in the main lobe direction is <math>D_m</math> m, and</li> <li>(iii) no other radio frequency (RF) sources with EIRP above 10 W are located within a distance of <math>5 D_m</math> m in the main lobe direction (as determined by considering the</li> </ul> |

<sup>2</sup> The criteria used for installation classes are based on [b-ICNIRP] limits. If other limits are used, the requirements in Table 8-1 may need to be adjusted.

<sup>3</sup> In no case can an EUT be deemed non-compliant based solely on its transmitted power.



**Table 8-1 – Example of product installation classes where a simplified evaluation process is applicable (based on [b-ICNIRP] general public limits)**

| Instal. class                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EIRP <sup>a</sup> (W) | EIRP <sup>a</sup> (dBm) | Installation criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |                         | half-power beam width) and within $D_m$ m in other directions. <sup>c</sup><br>$D_m$ is the compliance distance in the main lobe. If $D_m$ is not available, a value of 2 m can be used or 1 m if all EUT transmit frequencies are equal to or above 1 500 MHz.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| E+                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | > 100                 | > 50                    | The EUT is installed in accordance with instructions from the manufacturer and/or entity putting into operation and:<br>(i) the lowest radiating part of the antenna(s) is at a minimum height of $H_m$ m above the walking or standing surface accessible by the general public,<br>(ii) the minimum distance to areas accessible to the general public in the main lobe direction is $D_m$ m, and,<br>(iii) no other RF sources with EIRP above 100 W is located within a distance of $5D_m$ m in the main lobe direction and within $D_m$ m in other directions. <sup>d</sup><br>$D_m$ is the compliance distance in the main lobe. If $D_m$ is not available, it can be calculated using equations 8-1 to 8-4. $H_m$ is given by equations 8-1 to 8-4. |
| <sup>a</sup> EIRP transmitted by the installed antenna(s) including all its active bands.<br><sup>b</sup> According to [ITU-T K.52] emitters with a maximum EIRP of 2 W or less are inherently compliant.<br><sup>c</sup> When such condition is not fulfilled, the installation is still compliant if the sum of the EIRPs of the EUT and nearby sources is less than 100 W. If the total EIRP is above 100 W, then the EUT is still compliant if it is installed at a minimum height of $H_m$ m above the walking or standing surface accessible by the general public and at a minimum distance from areas accessible to the general public in the main lobe direction of $D_m$ m, where $H_m$ and $D_m$ are obtained using equations 8-1 to 8-4 for the sum of the EIRP values including those of nearby sources.<br><sup>d</sup> When such condition is not fulfilled, the installation is still exempted from measurements if the EUT is installed at a minimum height of $H_m$ m above the walking or standing surface accessible by the general public and at a minimum distance from areas accessible to the general public in the main lobe direction of $D_m$ m, where $H_m$ and $D_m$ are obtained using equations 8-1 to 8-4 for the sum of the EIRP values including those of nearby sources. |                       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

The criteria on the installation height for values of the EIRP  $\leq 100$  W were developed using the specific absorption rate (SAR) estimation formula provided in [IEC 62232] to ensure compliance with the basic restrictions. For an EIRP value  $> 100$  W,  $H_m$  and  $D_m$  (in metres) are given by equations 8-1 to 8-4.<sup>4</sup>

For frequencies between 100 MHz and 400 MHz:

$$H_m = \max \left\{ \begin{array}{l} 2 + \sqrt{\frac{EIRP \cdot A_{sl}}{2\pi}} \\ 2 + \sqrt{\frac{EIRP}{2\pi}} \sin(\alpha + 1.129\theta_{bw}) \end{array} \right. \quad D_m = \sqrt{\frac{EIRP}{2\pi}} \quad (8-1)$$

<sup>4</sup> For values of the EIRP  $\leq 100$  W, equations 8-1 to 8-4, based on a far-field condition, are not suitable.

For frequencies between 400 MHz and 2 000 MHz:

$$H_m = \max \begin{cases} 2 + \sqrt{\frac{EIRP \cdot 200 A_{sl}}{f\pi}} \\ 2 + \sqrt{\frac{200 \cdot EIRP}{f\pi}} \sin(\alpha + 1.129\theta_{bw}) \end{cases} \quad D_m = \sqrt{\frac{EIRP \cdot 200}{f\pi}} \quad (8-2)$$

For frequencies between 2 000 MHz and 300 000 MHz:

$$H_m = \max \begin{cases} 2 + \sqrt{\frac{EIRP \cdot A_{sl}}{10\pi}} \\ 2 + \sqrt{\frac{EIRP}{10\pi}} \sin(\alpha + 1.129\theta_{bw}) \end{cases} \quad D_m = \sqrt{\frac{EIRP}{10\pi}} \quad (8-3)$$

NOTE – Equations 8-1 to 8-4 are based on the reference levels in [b-ICNIRP] for general public exposure and reflect the fact that these are frequency dependent.

where:

$f$  is the frequency, in megahertz, of operation of the BS<sup>5</sup>

$A_{sl}$  is the side lobe suppression value<sup>6</sup>

$\alpha$  is the downtilt in radians (both electrical and mechanical)

$\theta_{bw}$  is the vertical half power beamwidth in radians.

Equations 8-1 to 8-3 have been obtained conservatively using the equations in the tables of Appendix III<sup>7</sup> of [ITU-T K.52]. In Figure 8-1 and Figure 8-2,  $D_m$  and  $H_m$  are given for some frequencies for the most stringent case based on a realistic choice of parameters for sector-coverage antennas ( $\theta_{bw} = \pi/12$ <sup>8</sup>,  $\alpha = \pi/12$ , and  $A_{sl} = 0.05$ <sup>9</sup>).

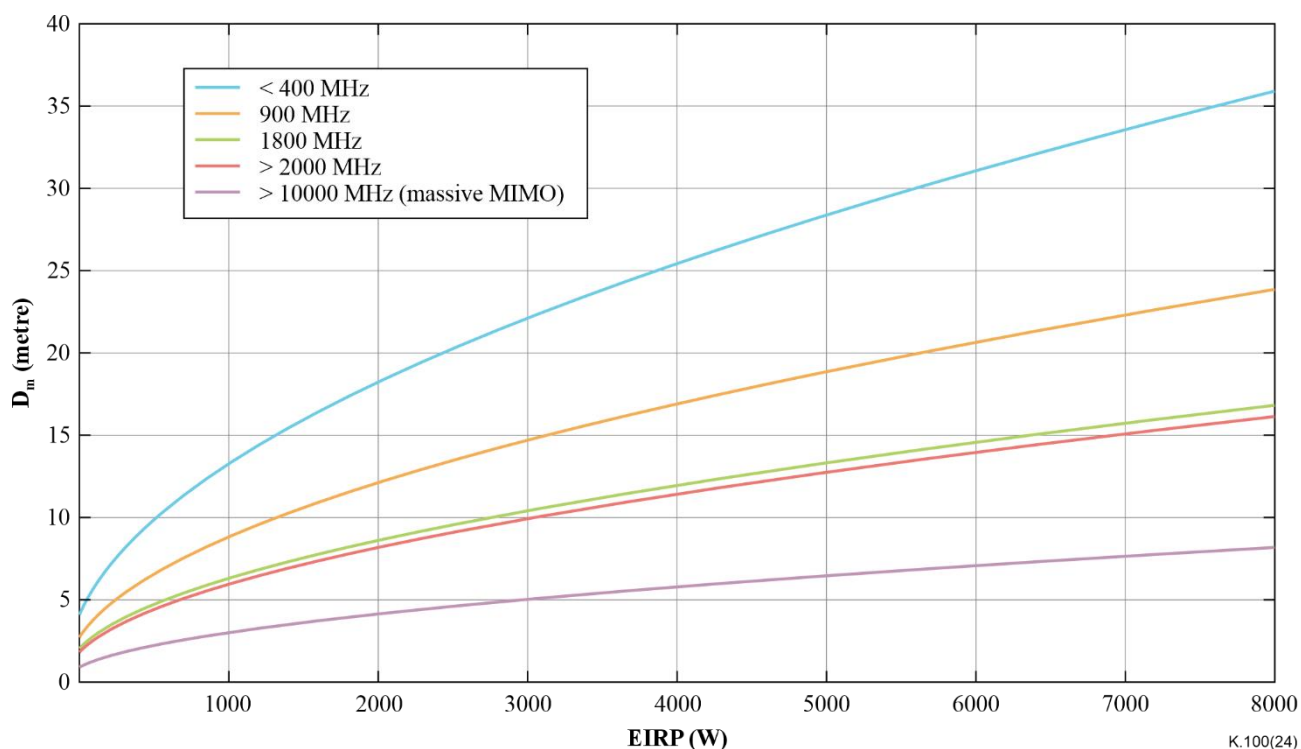
<sup>5</sup> To be conservative,  $f$  shall be chosen as the lowest limit in the frequency band of the EUT.

<sup>6</sup> Maximum side lobe amplitude with respect to the overall peak value.  $A_{sl}$  should be expressed as a numerical factor, however, it is usually given in decibels with respect to the maximum. To convert:  $A_{sl} = 10^{A_{sl}[\text{dB}]/10}$

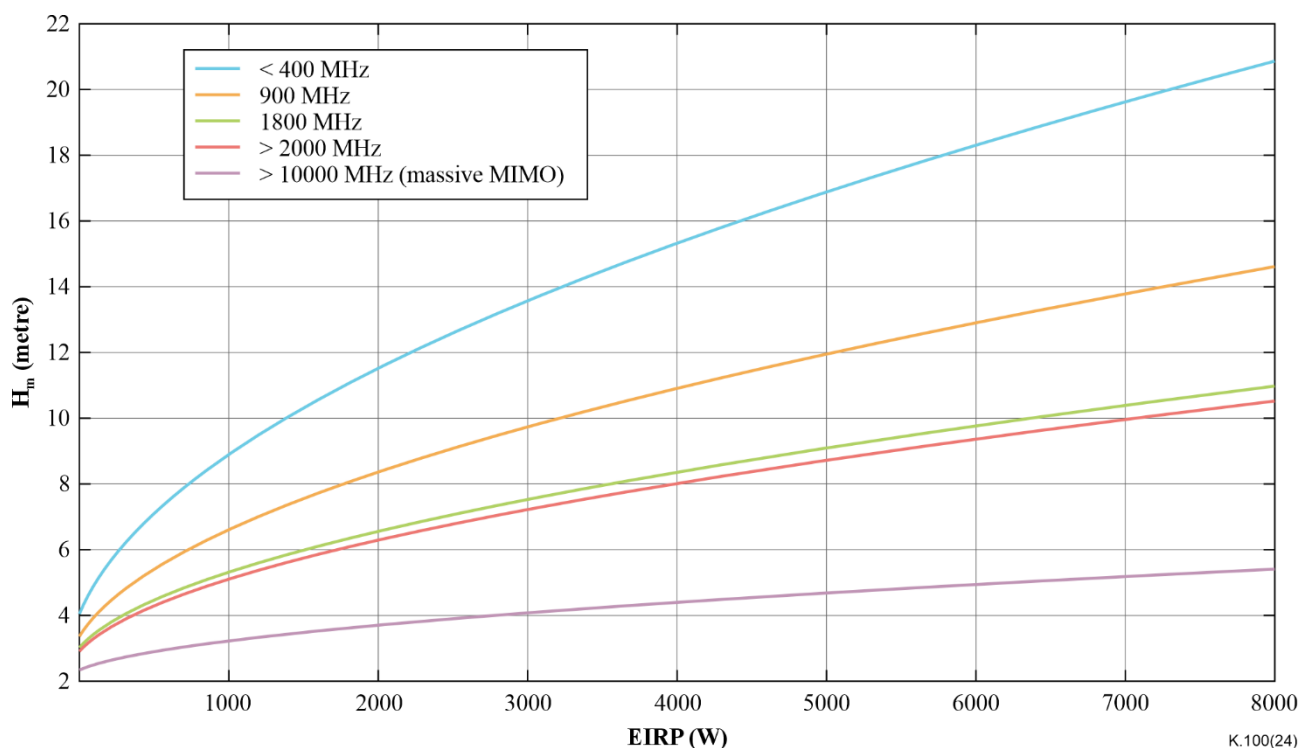
<sup>7</sup> Corresponding to directivity category 2 and accessibility category 1 of [ITU-T K.52] for  $H_m$  and accessibility category 2 for  $D_m$ .

<sup>8</sup>  $\pi/12$  corresponds to  $15^\circ$ .

<sup>9</sup> 0.05 corresponds to  $-13$  dB.



**Figure 8-1 –  $D_m$  as a function of the EIRP obtained from equations 8-1 to 8-3**



**Figure 8-2 –  $H_m$  as a function of the EIRP obtained from equations 8-1 to 8-3 based on a conservative choice of antenna parameters:  $\pi/12$  half-power beamwidth ( $\theta_{bw}$ ),  $\pi/12$  antenna downtilt ( $\alpha$ , electrical and mechanical) and side lobe suppression ( $A_{sl}$ ) of 0.05.  $H_m$  will decrease by decreasing  $\theta_{bw}$ ,  $\alpha$  and the side lobe suppression**

The criteria in Table 8-1 were developed to be applicable for a wide range of BS installations and provide general means to identify EUT exempted from measurements. Other sets of formulas

(e.g., those presented in [b-ANFR DR17] and [b-MSIP]) can be designed and used, e.g., for more specific installation types and exposure conditions, provided that their rationale and applicability are defined and documented. In general, when the impact of the environment and of ambient sources on the exposure of the EUT are known to be negligible or not relevant, the EUT is known to be compliant, without requiring further measurements, if the general public does not have access to its compliance boundary (CB). The CB of the EUT for the specific EUT configuration, if not already provided by the manufacturer, can be determined following procedures specified in [ITU-T K.61] and [IEC 62232].

Alternatively, in accordance with [IEC 62232], for BS using massive MIMO or beam steering antennas operating between 10 000 MHz and 300 000 MHz (i.e., 300 GHz) equation 8-4 applies:

$$H_m = \max \begin{cases} 2 + \sqrt{\frac{\text{EIRP} \cdot A_{sl}}{65\pi}} \\ 2 + \sqrt{\frac{\text{EIRP}}{65\pi}} \sin(\alpha + 1,129\theta_{3\text{dB}}) \end{cases} \quad \text{and} \quad D_m = \sqrt{\frac{\text{EIRP}}{40\pi}} \quad (8-4)$$

where symbols are the same as in equation 8-3.

## 9 Measurement procedure

### 9.1 Measurement area selection

The domain of investigation (DI) represents the area where a general exposure assessment (clause 9.4) shall be conducted. The DI is the part of the assessment domain boundary (ADB) of the EUT to which the general public has access.

Outside the ADB, the exposure ratio (ER) from the EUT is less than 0.05, i.e., the EUT is not a relevant source, and an assessment is not needed. The ADB for the EUT can be determined following procedures in [ITU-T K.61] and [IEC 62232]. Equation 9-1 is a simplified expression to obtain a conservative estimate of the ADB if the shape of a box (Figure 9-1) is provided.

$$L_{\text{ADB}} = \sqrt{\frac{\text{EIRP}}{0.05 \times 4\pi \cdot S_{\text{lim}}}} = 1.3 \times \sqrt{\frac{\text{EIRP}}{S_{\text{lim}}}} \quad (9-1)$$

where  $L_{\text{ADB}}$  is the side length of the ADB (in metres) in the main beam direction and  $S_{\text{lim}}$  is the relevant power density exposure limit (in watts per square metre). A plot of  $L_{\text{ADB}}$  as a function of the EIRP is provided in Figure 9-2 for some frequencies.

For multiband antennas having more than one active band (e.g., WCDMA 1900, LTE 800, NR 3500), the ADB may be calculated using equation 9-2:

$$L_{\text{ADB}} = 1.3 \times \sqrt{\sum \frac{\text{EIRP}_i}{S_{\text{lim}, i}}} \quad (9-2)$$

where  $\text{EIRP}_i$  is the EIRP of the EUT for band  $i$  and  $S_{\text{lim}, i}$  is the relevant power density exposure limit (in watts per square metre) for band  $i$ .

The dimensions of the ADB, if determined using equation 9-1 or equation 9-2, will be largely overestimated in the vertical direction of the antenna. Therefore, the following rules shall be applied.

- Regions placed  $H_b$  or more below the antenna mounting height (measured from the centre point of the antenna) shall not be considered as part of the ADB, where  $H_b$  (in metres) is given by:<sup>10</sup>

$$H_b = \max(L_{ADB} \cdot \tan \alpha; 3.5)$$

Here,  $\alpha$  is the antenna downtilt (mechanical and electrical) in radians. If  $\alpha$  is not known it can be conservatively assumed to equal to  $\pi/15$ .

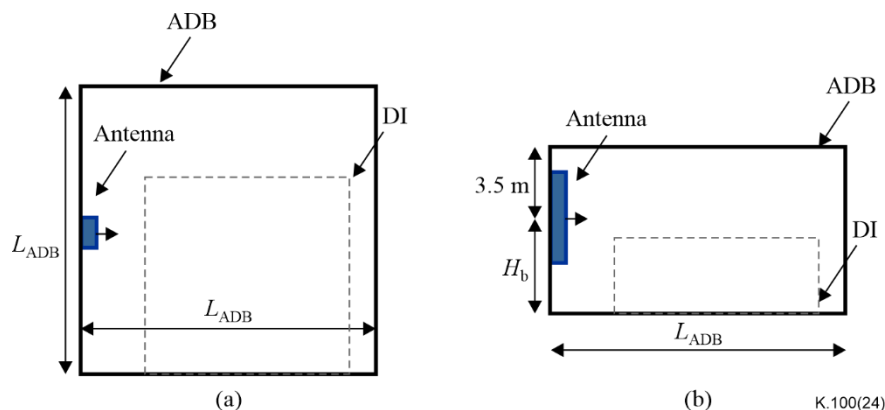
- Regions placed  $\geq 3.5$  m above the antenna mounting height (measured from the centre point of the antenna) shall not be considered as part of the ADB.

Equations 9-1 and 9-2 apply for downtilted antennas. If the antenna is tilted upward, the values shall be exchanged.

In addition, for rooftop or wall installations, regions within the building on which the antenna is mounted shall be excluded from the ADB if the antenna main beam is pointing away from the building.<sup>11</sup>

Based on observations of the EUT installation and environment, calculation by means of numerical tools (such as the EMF estimator [ITU-T K.70]), as well as on experience gained by assessments of similar sites, the DI can be restricted only to those points where the level of exposure is expected to be relevant or maximum (see, for instance, Appendix III).

If the general public has no access to the ADB, there is no DI and the EUT is compliant, i.e., no measurements are needed.

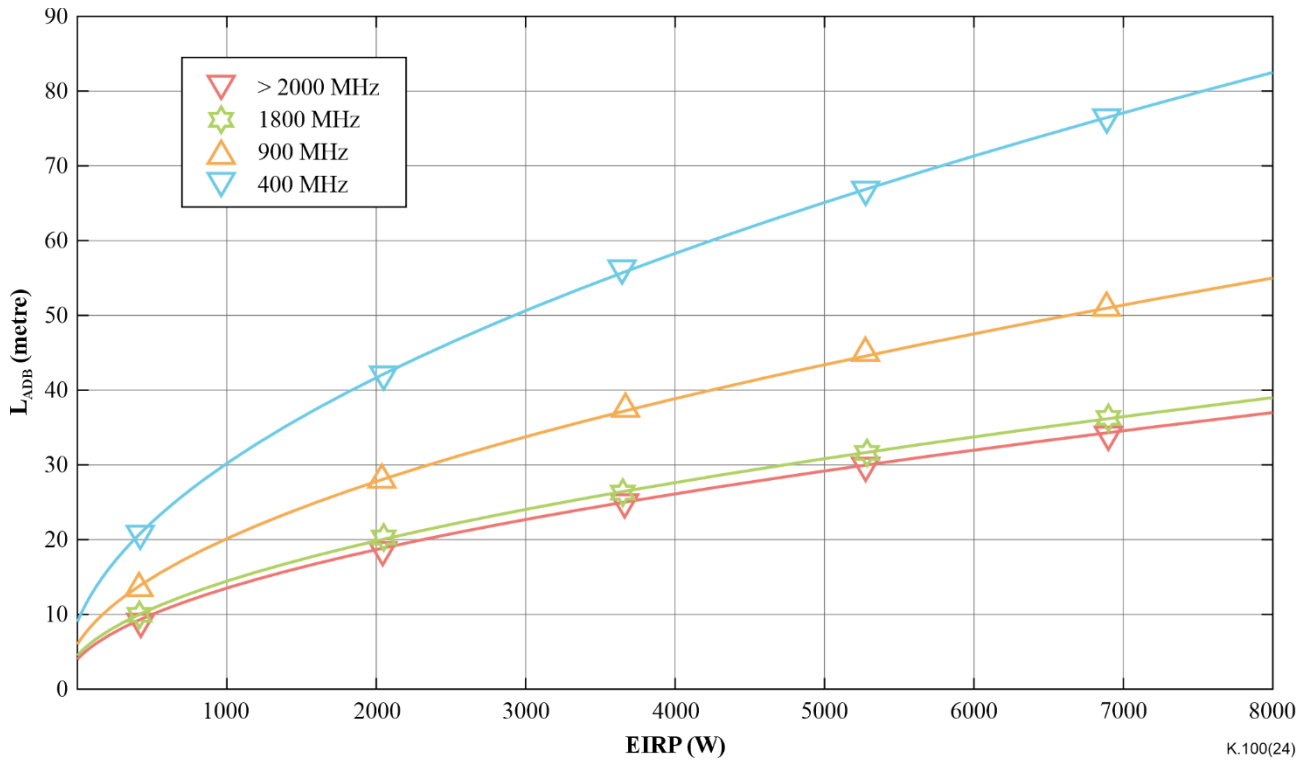


**Figure 9-1 – a) Top-view of the square-shaped assessment domain boundary (ADB) with side length  $L_{ADB}$**

**The ADB is oriented according to the antenna direction**  
**b) Side-view of the ADB determined according to clause 9**

<sup>10</sup> The first term in brackets corresponds to the height of the ADB given by the main beam for downtilted antennas. The second term takes into consideration that, for small tilt angles, the ADB in the vertical direction might be given by the antenna side lobe (or by the extension of the main lobe in the vertical plane). Since the maximum side lobe amplitude and direction might be difficult to estimate, a minimum height of 3.5 m is conservatively chosen.

<sup>11</sup> Transmission in these directions corresponds to the side lobe of the antenna. In addition, the attenuation in the walls and roof can reduce the power density by 10-20 dB or more.



**Figure 9-2 – Side length of the assessment domain boundary as a function of the EIRP for some frequencies, estimated using equation 9-1<sup>12</sup>**

## 9.2 Measurement equipment

Frequency-selective or broadband measurement equipment can be used to assess compliance of the EUT. Broadband measurements provide directly the sum of all power density values from signals over the frequency range of the probe without distinguishing the contribution of different frequencies (whether from the EUT or from ambient sources). Broadband measurement results may be extrapolated to estimate the maximum possible RF field strength (see clause 9.5). Such extrapolation, however, can result in a vast overestimation depending on the characteristic of the probe and the characteristics of the EUT and the ambient signals. Therefore, frequency-selective measurements are recommended where accurate extrapolation is required.

For broadband and frequency-selective equipment, the RF field strength measurement shall consider contributions from all directions and polarizations. An isotropic probe is best suited for this, but other antennas may be used. For instance, a single-axis probe (e.g., dipole) can be used by positioning the probe in three orthogonal directions and summing the individual contributions,  $E_N$ , so that:

$$E = \sqrt{\sum_{N=1}^{N=3} E_N^2}$$

Additional requirements on the measurement equipment are provided in Appendix I and in [IEC 62232].

<sup>12</sup> Curves are based on [b-ICNIRP] limits.

### 9.3 Measured quantity

At sufficiently large distances from the source, only the r.m.s. electric ( $E$ ) or magnetic ( $H$ ) field strength need to be measured and the ER can be calculated as:

$$ER = \frac{E^2}{E_{\text{lim}}^2} \cong \frac{H^2}{H_{\text{lim}}^2} \cong \frac{S}{S_{\text{lim}}}$$

where  $S$  is the plane wave equivalent power density ( $S = E^2 / \eta_0 = \eta_0 H^2$  for  $\eta_0 \approx 377 \Omega$ ). This relation is verified already outside the reactive near-field region of the antenna [IEC 62232], where measurements are typically conducted.

When exposure is evaluated for a certain frequency band (the total power density or the field strength within the frequency interval  $[f_{\text{min}}, f_{\text{max}}]$  is assessed),  $S_{\text{lim}}$ ,  $E_{\text{lim}}$  or  $H_{\text{lim}}$  are chosen as the most stringent limits within the band.

The relevant exposure standard may specify the applicable time averaging period relevant for the field strength and power density measurements (e.g., any 6 min in [b-ICNIRP]). For general exposure assessment of the instantaneous exposure level, time averaging over other periods may be acceptable (r.m.s. field strength values averaged over a shorter time can be used, see [ITU-T K.156] [b-Kim, 2010]).

### 9.4 General TER measurement

The general exposure assessment consists of measurement of the total field strength over the entire frequency range covered by the Recommendation or at least in the range of frequencies used by the technologies present on site, including EUT and ambient sources. Broadband equipment is therefore suitable for this type of measurement. Frequency-selective instruments can also be used by integrating the field strength over the entire bandwidth.

No further assessments are required, and the site is deemed compliant, if the maximum measured TER in DI is lower than 0.05. Otherwise, the location(s) where the maximum level of exposure was found shall be selected for comprehensive measurements (see clause 9.5).

The application of the condition above is recommended since it provides an efficient tool to show compliance of the EUT with simple broadband measurements. Despite the total exposure ratio (TER) value measured during general exposure assessment the user of this guide may always choose to conduct a comprehensive assessment (see dotted line in the flowchart of Figure 6-1).

During general exposure assessments, measurements should be taken 1.5 m above the ground. Good practice is to move the probe slowly through the DI since probes generally are sensitive to fast movements.

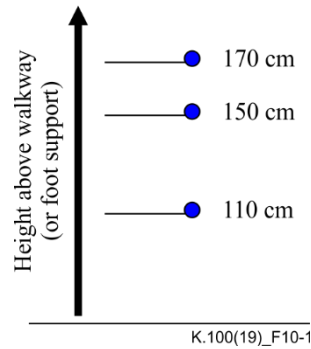
### 9.5 Comprehensive EUT ER measurement

The comprehensive EUT ER measurement is conducted at the location(s) of maximum TER identified in the general TER evaluation. It takes into account time averaging in the same way as TER. It is performed to obtain a conservative estimate of the exposure level of the EUT, corresponding to the ER of the EUT when the product is transmitting at configured maximum power. The assessment is performed using broadband or frequency-selective measurement equipment, see clauses 9.5.1 and 9.5.2 respectively. For this purpose, broadband measurements generally result in a large overestimation of the maximum RF field strength and the extent of the overestimation shall be considered and reported. For comprehensive measurements, the extrapolation procedure as described in clause 9.5.1 shall be applied.

During measurements the distance between the measurement equipment and reflecting objects should be at least 1 m.

Each measurement point should be assessed at three heights: 1.1 m, 1.5 m, and 1.7 m (see Figure 10-1).<sup>13</sup> Among those, the largest value reported shall be used for comparison with the exposure limit.

No further assessments are required, and the site is deemed compliant if the ER of the EUT obtained with comprehensive measurements is less than 0.05.



**Figure 10-1 – Location of measurement points**

### 9.5.1 Comprehensive assessment with frequency-selective equipment

Extrapolation is applicable when the measurement has been performed at EUT power levels lower than the maximum transmitted power. Extrapolation can be used, for example, to determine the maximum or actual maximum power density or RF field strength values from the measurement of technology-specific stable (i.e., time independent) reference signals (e.g., pilot channels).

To measure such reference signal, frequency-selective equipment and in some cases a specific decoder is needed (see Appendix III and [IEC 62232]).

The time-invariant signals used for different technologies are, for example:

- the BCCH for GSM;
- the CPICH for (UMTS);
- the primary broadcast channel, physical broadcast channel (PBCH) or the reference signal, RS, for LTE;
- the synchronisation signal block (SSB) or channel state information reference signal (CSI-RS) for 5G NR.

The ratio between maximum possible signal and the time invariant component of the signal is determined based on knowledge of the technology and the specific BS configuration. This ratio corresponds to the extrapolation factor,  $F_{\text{ext}}$ :

$$S_{\text{max}} = S_{\text{narrow}} \cdot F_{\text{ext}}$$

where  $S_{\text{narrow}}$  is the measured power density for the time invariant component of the signal and  $S_{\text{max}}$  the corresponding value that would be measured if the BS transmitted at theoretical maximum power.

Note that the extrapolation factors are valid for power density values. For electric and magnetic field strength levels, the square root of the extrapolation factor for power density shall be used:

$$E_{\text{max}} = E_{\text{narrow}} \cdot \sqrt{F_{\text{ext}}}$$

The values of  $F_{\text{ext}}$  for different technologies are provided in Appendix II together with the recommended measurement settings.

<sup>13</sup> When justified, measurements at other heights may be conducted.



If the equipment is able to implement the actual maximum approach, extrapolation shall be performed for the actual maximum RF field strength or power density. This shall be considered when deriving the relevant extrapolation factor as described in [IEC 62232], for example by scaling the extrapolation factor ( $F_{\text{ext}}$ ) by the power reduction factor ( $F_{\text{PR}}$ ).

### 9.5.2 Comprehensive assessment with broadband equipment

When a broadband equipment is used for comprehensive assessments, the measurement data can be extrapolated to the maximum or actual maximum transmitted power as described in clause 9.5.1 only when the following conditions are satisfied.

- a) The EUT transmissions are in a single (known) frequency carrier/channel/band.
- b) There are no relevant ambient signals.
- c) The EUT parameters at the time of measurement are known (especially the transmitted power).
- d) The highest extrapolation factor among all technologies available in the frequency band of the EUT is applied, see [IEC 62232].

When evaluating the exposure ratio, if the applicable RF exposure limit is frequency dependent, the lowest exposure limit value for the frequency range of the EUT is applied.

Extrapolation performed under these conditions, results in an overestimation of the maximum RF field strength. The extent of the overestimation should be evaluated and reported.

If the extrapolated result exceeds the exposure limit, it does not necessarily mean that the limits are exceeded. For such a case, extrapolation based on frequency selective measurements, see clause 9.5.1 should be used to verify compliance with the limit.

## 9.6 Exposure contribution of ambient sources

If extrapolation of the field strength to the maximum power has been obtained by means of broadband measurements, the contribution from ambient sources is implicitly and conservatively assessed and no additional measurements need to be conducted.

If a frequency-selective instrument has been used for comprehensive measurements of the EUT exposure and the resulting ER is equal to or larger than 0.05, it shall be identified whether RF sources other than the EUT are to be considered as relevant. Each RF source (other than the EUT) with a measured (non-extrapolated)  $ER \geq 0.05$  is considered as a relevant source. Sources with lower ER values should be excluded.

If the operating bands of nearby ambient sources are known, the ER contribution of each source is measured with the frequency-selective equipment by integrating the field strength over the corresponding band. When this information is not directly available, it can be retrieved by inspection of the significant peaks in the spectrum.

The field strength from relevant source contributions shall be scaled to maximum power in a manner similar to the description in clause 9.5. If the extrapolation factors are unknown and cannot be recovered or the time invariant signals used as reference are not accessible, the power density for each of the relevant bands shall be measured during high-traffic hours using a max-hold trace in order to store the maximum value of all measurement values until the equipment reading stabilizes (typically 1 min or less).

For broadcasting systems such as frequency modulation (FM) radio, digital audio broadcasting (DAB), digital radio and digital video broadcasting-terrestrial (DVB T) extrapolation is not needed, since their transmitted power is typically time invariant.

## 9.7 Determination of the total exposure ratio

At the measurement points of the DI where a general exposure assessment has reported maximum exposure level, the total exposure ratio (TER) shall be calculated using the following formula.

$$TER = \sum_{i=1}^N ER_i$$

where  $ER_i$  is the ER for the frequency band  $i$ .

## 10 Uncertainty

Uncertainty shall be estimated for every reported measured and calculated RF field strength, power density or SAR evaluation. Guidance on how to estimate uncertainty can be found in [IEC 62232] and [b-JCGM 100]. Although [b-JCGM 100] is concerned with measurement uncertainty, the same concepts can be extended to computational uncertainty. The measurement uncertainty should be obtained including the sources of uncertainty identified in Table 10-1 or Table 10-2, see [IEC 62232], and in [ITU-T K.91]. A description of the uncertainty factors can be found in [IEC 62232].

The target expanded uncertainty is 4 dB or below, which is considered industry best practice. The expanded uncertainty for the RF exposure evaluation used for the product installation compliance assessment should not exceed 6 dB. Under this condition, the evaluation result shall be compared directly to the exposure limit. However, the entity performing the uncertainty analysis also needs to be aware of the applicable regulation.

**Table 10-1 – Sample template for estimating the expanded uncertainty of a radio frequency field strength measurement that used a frequency-selective instrument [IEC 62232]**

| Source of uncertainty<br>(influence quantity)                     | Unit | Prob-<br>ability<br>distrib-<br>ution<br>type | Uncertainty<br>or semi span<br>$a$ | Divisor<br>$d$ | Sensi-<br>tivity<br>coeffi-<br>cient<br>$c$ | Standard<br>uncertainty<br>$u = a/d$ | Corr.<br>fact<br>$t$ | $c^2u^2$ |
|-------------------------------------------------------------------|------|-----------------------------------------------|------------------------------------|----------------|---------------------------------------------|--------------------------------------|----------------------|----------|
| <b>Measurement equipment</b>                                      |      |                                               |                                    |                |                                             |                                      |                      |          |
| Calibration of the meter (or spectrum analyser)                   | dB   | normal                                        |                                    | 1.96           | 1                                           |                                      |                      |          |
| Calibration of the antenna factor                                 | dB   | normal                                        |                                    | 1.96           | 1                                           |                                      |                      |          |
| Calibration of the cable loss                                     | dB   | normal                                        |                                    | 1.96           | 1                                           |                                      |                      |          |
| Combined frequency response of the meter/cable/antenna            | dB   | rect.                                         |                                    | $\sqrt{3}$     | 1                                           |                                      |                      |          |
| Combined linearity deviation of the meter/cable/antenna           | dB   | rect.                                         |                                    | $\sqrt{3}$     | 1                                           |                                      |                      |          |
| Isotropy of the antenna                                           | dB   | rect.                                         |                                    | $\sqrt{3}$     | 1                                           |                                      |                      |          |
| Combined temperature and humidity response of meter/cable/antenna | dB   | rect.                                         |                                    | $\sqrt{3}$     | 1                                           |                                      |                      |          |

**Table 10-1 – Sample template for estimating the expanded uncertainty of a radio frequency field strength measurement that used a frequency-selective instrument [IEC 62232]**

| Source of uncertainty<br>(influence quantity)                                                 | Unit | Prob-<br>ability<br>distrib-<br>ution<br>type | Uncertainty<br>or semi span<br>$a$ | Divisor<br>$d$ | Sensi-<br>tivity<br>coeffi-<br>cient<br>$c$ | Standard<br>uncertainty<br>$u = a/d$ | Corr.<br>fact<br>$t$ | $c^2u^2$ |
|-----------------------------------------------------------------------------------------------|------|-----------------------------------------------|------------------------------------|----------------|---------------------------------------------|--------------------------------------|----------------------|----------|
| Mismatch between antenna and meter/spectrum analyser                                          | dB   | U                                             |                                    | $\sqrt{2}$     | 1                                           |                                      |                      |          |
| <b>Methodology</b>                                                                            |      |                                               |                                    |                |                                             |                                      |                      |          |
| Field scattering from surveyor's body                                                         | dB   | rect.                                         |                                    | $\sqrt{3}$     | 1                                           |                                      |                      |          |
| Mutual coupling between measurement antenna or isotropic probe and object                     | dB   | rect.                                         |                                    | $\sqrt{3}$     | 1                                           |                                      |                      |          |
| <b>Source and environment</b>                                                                 |      |                                               |                                    |                |                                             |                                      |                      |          |
| Field reflections from movable large objects near the source during measurement               | dB   | rect.                                         |                                    | $\sqrt{3}$     | 1                                           |                                      |                      |          |
| Scattering from nearby objects and the ground                                                 | dB   | rect.                                         |                                    | $\sqrt{3}$     | 1                                           |                                      |                      |          |
| Combined correction factor, $t_c = \sum_{i=1}^N t_i$                                          |      |                                               |                                    |                |                                             |                                      |                      | N/A      |
| Combined standard uncertainty, $u_c = \sqrt{\sum_{i=1}^N (c_i^2 u_i^2)}$                      |      |                                               |                                    |                |                                             |                                      |                      |          |
| Coverage factor for required (e.g., 95 %) confidence interval, $k$                            |      |                                               |                                    |                |                                             |                                      |                      |          |
| Expanded uncertainty, $U = k \times u_c$                                                      |      |                                               |                                    |                |                                             |                                      |                      |          |
| NOTE 1 – The value of divisor $d$ for normal probability distribution is for 95 % confidence. |      |                                               |                                    |                |                                             |                                      |                      |          |
| NOTE 2 – See [IEC 62232] for guidance on the variables in this table.                         |      |                                               |                                    |                |                                             |                                      |                      |          |

**Table 10-2 – Sample template for estimating the expanded uncertainty of a radio frequency field strength measurement that used a broadband instrument [IEC 62232]**

| Source of uncertainty (influence quantity)                                                        | Unit | Probability distribution type | Uncertainty or semi span $a$ | Divisor $d$ | Sensitivity coefficient $c$ | Standard uncertainty $u = a/d$ | Correlation factor $t$ | $c^2 u^2$ |
|---------------------------------------------------------------------------------------------------|------|-------------------------------|------------------------------|-------------|-----------------------------|--------------------------------|------------------------|-----------|
| <b>Measurement equipment</b>                                                                      |      |                               |                              |             |                             |                                |                        |           |
| Calibration of field probe                                                                        | dB   | normal                        |                              | 1.96        | 1                           |                                |                        |           |
| Frequency response of field probe                                                                 | dB   | rect.                         |                              | $\sqrt{3}$  | 1                           |                                |                        |           |
| Isotropy of the field probe                                                                       | dB   | rect.                         |                              | $\sqrt{3}$  | 1                           |                                |                        |           |
| Temperature response of the field probe                                                           | dB   | rect.                         |                              | $\sqrt{3}$  | 1                           |                                |                        |           |
| Linearity deviation of the field probe                                                            | dB   | rect.                         |                              | $\sqrt{3}$  | 1                           |                                |                        |           |
| <b>Methodology</b>                                                                                |      |                               |                              |             |                             |                                |                        |           |
| Field reflections from surveyor's body                                                            | dB   | rect.                         |                              | $\sqrt{3}$  | 1                           |                                |                        |           |
| Mutual coupling between measurement antenna or isotropic probe and object                         | dB   | rect.                         |                              | $\sqrt{3}$  | 1                           |                                |                        |           |
| <b>Source and environment</b>                                                                     |      |                               |                              |             |                             |                                |                        |           |
| Scattering from nearby objects and the ground                                                     | dB   | rect.                         |                              | $\sqrt{3}$  | 1                           |                                |                        |           |
| Field reflections from movable large objects near the source                                      | dB   | rect.                         |                              | $\sqrt{3}$  | 1                           |                                |                        |           |
| Combined correction factor, $t_c = \sum_{i=1}^N t_i$                                              |      |                               |                              |             |                             |                                |                        | N/A       |
| Combined standard uncertainty, $u_c = \sqrt{\sum_{i=1}^N (c_i^2 u_i^2)}$                          |      |                               |                              |             |                             |                                |                        |           |
| Coverage factor for required (e.g., 95 %) confidence interval, $k$                                |      |                               |                              |             |                             |                                |                        |           |
| Expanded uncertainty, $U = k \times u_c$                                                          |      |                               |                              |             |                             |                                |                        |           |
| NOTE 1 – The value of divisor $d$ for the normal probability distribution is for 95 % confidence. |      |                               |                              |             |                             |                                |                        |           |
| NOTE 2 – See [IEC 62232] for guidance on the variables in this Table.                             |      |                               |                              |             |                             |                                |                        |           |

An example where uncertainty estimation for some of the factors listed in Tables 11-1 and 11-2 is described in [b-Kim, 2012].

## Appendix I

### Radio frequency field strength measurement equipment requirements

(This appendix does not form an integral part of this Recommendation.)

The measurement equipment should be calibrated at a sufficient number of frequencies to achieve the declared uncertainty of the equipment over the measurement frequency range. Table I.1 summarizes the performance requirements for a broadband measurement system [IEC 62232].

**Table I.1 – Broadband measurement system requirements**

| Parameters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Requirements                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frequency response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>±1.5 dB (from 600 MHz to 6 GHz)</li> <li>±3 dB (for the frequencies to be measured &lt; 600 MHz and &gt; 6 GHz)</li> </ul>                                                    |
| <b>Lower detection level<sup>a</sup></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>17 dB below the lowest value specified in the applicable exposure limits or regulation in the frequency range covering the EUT and ambient sources, see clause 9.6</li> </ul> |
| <b>Dynamic range</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | > 25 dB                                                                                                                                                                                                              |
| <b>Linearity</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ±1.5 dB                                                                                                                                                                                                              |
| <b>Probe isotropy<sup>b</sup></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | < 2.5 dB for an isotropic probe                                                                                                                                                                                      |
| <p>a This allows to determine contributions of relevant sources, (i.e., 13 dB range considering an ER of 0.05) including a 4 dB margin corresponding to the target uncertainty, see clause 10.</p> <p>b Probes and measurement antennas with isotropic response are recommended. Single-axis (e.g., dipole) and directional measurement antennas are permitted provided that the measurements are post processed to obtain the total RF field strength (equivalent to a measurement with an isotropic probe or measurement antenna).</p> |                                                                                                                                                                                                                      |

Table I.2 summarizes the performance requirements for the frequency-selective measurement system [IEC 62232].

**Table I.2 – Frequency-selective measurement system requirements**

| Parameters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Requirements                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frequency response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>±1.5 dB (from 600 MHz to 6 GHz)</li> <li>±3 dB (for the frequencies to be measured &lt; 600 MHz and &gt; 6 GHz)</li> </ul>                                                                                                                                   |
| <b>Lower detection level<sup>a</sup></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>44 dB below the lowest value specified in the applicable exposure limits or regulation in the frequency range covering the EUT and ambient sources, see clause 9.6.</li> <li>Signal-to-noise ratio of at least 10 dB in the measurement bandwidth</li> </ul> |
| <b>Dynamic range</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | > 60 dB                                                                                                                                                                                                                                                                                             |
| <b>Linearity</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ±1.5 dB                                                                                                                                                                                                                                                                                             |
| <b>Probe isotropy<sup>b</sup></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>&lt; 2 dB (&lt; 900 MHz)</li> <li>&lt; 3 dB (from 900 MHz to 3 GHz)</li> <li>&lt; 5 dB (&gt; 3 GHz)</li> </ul>                                                                                                                                               |
| <p>a This allows to determine contributions of sources of interest (i.e., 40 dB range considering an ER of 0.0001) including a 4 dB margin corresponding to the target uncertainty, see clause 10.</p> <p>b Probes and measurement antennas with isotropic response are recommended. Single-axis (e.g., dipole) and directional measurement antennas are permitted provided that the measurements are post processed to obtain the total RF field strength (equivalent to a measurement with an isotropic probe or measurement antenna).</p> |                                                                                                                                                                                                                                                                                                     |

## Appendix II

### Guidance on comprehensive measurements for specific technologies

(This appendix does not form an integral part of this Recommendation.)

This appendix provides guidance on the spectrum analyser settings required to measure signals from different technologies and to extrapolate them to the maximum emitted power condition. Accurate measurements with a spectrum analyser require the settings of parameters such as:

- detection mode;
- resolution bandwidth (RBW);
- frequency span or central frequency ( $f_{\text{cent}}$ ).

#### II.1 Time division multiple access/frequency division multiple access technology

Time division multiple access (TDMA) mobile phone technology (e.g., GSM or TETRA) and frequency division multiple access (FDMA) mobile phone technology (e.g., TETRAPOL) utilize a time invariant BS radio channel that operates at constant full power and can be used as a stable reference.

For example, in the GSM system, this constant power channel is known as the BCCH. Table II.1 lists constant power components for various technologies.

**Table II.1 – Example constant power components for specific technologies**

| Technology | Constant power component         |
|------------|----------------------------------|
| GSM        | BCCH                             |
| TETRA      | Multicast control channel (MCCH) |
| TETRAPOL   | MCCH                             |

If the traffic channels each operate at a maximum power equal to the constant power component, which is the case for GSM, then a conservative maximum transmit power,  $P_{\text{max}}$ , can be determined by multiplying the power of the constant power component,  $P_{\text{const}}$ , by the total number of radio channels that feed into the antenna,  $N_{\text{GSM}}$ . Therefore, the power density corresponding to the maximum emitted power condition,  $S_{\text{max}}$ , can be obtained by measuring the power density in the BCCH ( $S_{\text{BCCH}}$ ) of the EUT scaled by  $N_{\text{GSM}}$ :

$$S_{\text{max}} = S_{\text{BCCH}} \cdot F_{\text{ext\_GSM}}$$

For measuring the BCCH, the following settings for the frequency-selective equipment are recommended:

- $f_{\text{cent}}$ : BCCH central frequency;
- RBW: 200 kHz (smaller RBW can be used as long as all the contributions in the occupied bandwidth of the BCCH signal are summed, higher RBW would include the power of adjacent channels);
- detection mode: r.m.s.

#### II.2 Code division multiple access/wideband code division multiple access technology

Code division multiple access code division multiple access (CDMA)/WCDMA mobile phone systems use spread spectrum technology employing a constant power control or pilot channel that has a fixed power relationship to the maximum allocated power. Dedicated decoders are available that

enable the constant power reference channel (e.g., CPICH in UMTS/WCDMA) to be measured allowing calculation of maximum RF field strength.

If the ratio of the maximum allocated power to the power in the control channel of the EUT is  $F_{\text{extCPICH}}$  and the measured RF power density from the control channel is  $S_{\text{CPICH}}$  then the extrapolated value is:

$$S_{\text{max}} = S_{\text{CPICH}} \cdot F_{\text{ext\_CPICH}}$$

The parameter  $N_{\text{CPICH}}$  is set by the telecommunications operator. A typical value is 10 (i.e., 10% of total power allocated to CPICH).

### II.3 ODFM technology

For LTE, which use orthogonal frequency-division multiplexing (OFDM) technology, two types of reproducible methods are described: one method requires the use of a dedicated decoder (similar to existing methods based on pilot signals as for WCDMA) and another method for which a decoder is not needed. For NR, more details are provided in [IEC 62232].

#### *Method using a dedicated decoder*

By means of an LTE decoder, the reference signal RS, transmitted by the BS at a constant power level, is measured and extrapolated to the maximum power density according to the following expression:

$$S_{\text{max}} = \frac{F_{\text{ext\_RS}}}{BF} \cdot (S_{\text{RS\_PORT1}} + S_{\text{RS\_PORT2}} + \dots + S_{\text{RS\_PORTn}})$$

where  $S_{\text{RS\_PORT1}}, S_{\text{RS\_PORT2}} \dots S_{\text{RS\_PORTn}}$  are the measured power density values of the reference signal (RS)<sup>14</sup> transmitted by each antenna port,  $N_{\text{RS}}$  is the ratio of the BS maximum power to the power of the RS and  $BF$  is the power boosting factor.<sup>15</sup>  $F_{\text{ext\_RS}}$  corresponds to the number of subcarriers and is dependent by the LTE channel bandwidth, see Table II.2.

**Table II.2 – Theoretical extrapolation factor,  $N_{\text{RS}}$**

| Bandwidth<br>[MHz] | $F_{\text{ext\_RS}}$ (linear/dB) |
|--------------------|----------------------------------|
| 1.4                | 72/18.57                         |
| 3                  | 180/22.55                        |
| 5                  | 300/24.77                        |
| 10                 | 600/27.78                        |
| 15                 | 900/29.54                        |
| 20                 | 1 200/30.79                      |

Since the RSs are uniformly distributed over the occupied radio bandwidth, this method is recommended in environments with strong selective fading.

<sup>14</sup> The RS field strength is measured as the linear average over the field strength contributions of all resource elements that carry the RS within the operating bandwidth. Thus, the measured value corresponds to the average power transmitted for one 15 kHz subcarrier.

<sup>15</sup> The RS can be transmitted from either one, two or four antennas (or antenna ports). The instrument should be able to determine the RS power for all antennas separately. The BF value can be obtained by the operator.

### Method using a spectrum analyser

A basic spectrum analyser is less expensive and more commonly available compared with a dedicated LTE decoder. However, the powers of the RSs cannot be accurately detected, since they are transmitted on single resource elements spread in frequency and time.

To overcome this issue and to avoid requirements on access to *a priori* knowledge regarding band occupation or service characteristics, the PBCH power can be measured. The PBCH is transmitted with the same characteristics regardless of the configuration or service bandwidth and spans a bandwidth of approximately 1 MHz over the centre frequency of the LTE signal.

Please note that the signal from each LTE BS cannot be identified using this method due to frequency spectrum overlapping.

The measured peak power density,  $S_{\text{PBCH}}$ , corresponds to the received PBCH signal power over the bandwidth of 72 subcarriers (each of 15 kHz). The maximum power density,  $S_{\text{max}}$ , of the LTE signal at each measurement point is then given by:

$$S_{\text{max}} = S_{\text{PBCH}} \cdot F_{\text{ext\_PBCH}}$$

where  $F_{\text{ext\_PBCH}}$  is the extrapolation factor for the PBCH, which is the ratio of the maximum transmission power to the transmission power corresponding to the PBCH over six resource blocks.  $N_{\text{PBCH}}$  can be calculated theoretically according to:

$$F_{\text{ext\_PBCH}} = \frac{F_{\text{ext\_RS}}}{72}$$

where  $N_{\text{RS}}$  denotes the number of subcarriers in the transmission bandwidth used, see Table II.2.

To measure the PBCH, the following settings for the frequency-selective equipment are recommended:

- $f_{\text{cent}}$ : central frequency LTE signal;
- RBW: 1 MHz (smaller RBW can be used as long as all the contributions in the bandwidth of the PBCH signal occupied are summed);
- detection mode: r.m.s.;
- frequency span set to zero (scope mode);
- sweep time:  $70 \mu\text{s} \cdot \text{SA}_{\text{points}}$ , where  $\text{SA}_{\text{points}}$  is the number of display points of the spectrum analyser (this is done in order to obtain an integration time close to the symbol duration of each pixel on the screen of the spectrum analyser);
- minimum 20 s sweep time and peak trace of the power.



## Appendix III

### Location of the point of maximum exposure (POME) for a base station in the line of sight

(This appendix does not form an integral part of this Recommendation.)

The probable location of the point of maximum exposure (POME) to EMF associated with an antenna main lobe for line-of-sight scenarios (see Figure III.1) can be estimated using the following equation [b-Linhares]:

$$X_{\max}^{\text{exp}} = \frac{\sqrt{\left[ (H - h) \tan \alpha \left( 1 + \frac{q}{2} \right) \right]^2 + 2q(H - h)^2} - (H - h) \tan \alpha \left( 1 + \frac{q}{2} \right)}{2}$$

where  $X_{\max}^{\text{exp}}$  (in metres) indicates the estimated horizontal distance from the antenna in the direction of the main lobe where maximum exposure is expected,  $H$  is the antenna height (in metres) from ground to the centre of the antenna,  $h$  is the reference point height at which measurements should be conducted (in metres),  $\alpha$  is the antenna downtilt (mechanical and electrical) and  $q$  is calculated by:

$$q = \frac{\log(0.5)}{\log \left[ \cos \left( \frac{\theta_{bw}}{2} \right) \right]}$$

where  $\theta_{bw}$  is the antenna half-power beamwidth.

The estimated location  $X_{\max}^{\text{exp}}$  indicates the probable region of maximum exposure. Nevertheless, it must be taken into account that scattered and diffracted fields may shift the real local maximum in other regions. Moreover, due to the presence of other relevant sources, the largest exposure level may be outside the antenna main beam direction of the EUT. In general, it is therefore recommended to sweep the instruments in the DI (see clause 9) to locate the point(s) of maximum exposure.

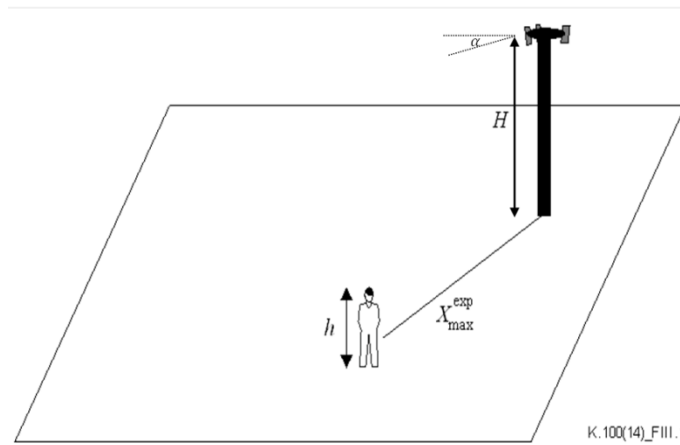


Figure III.1 – Geometry set-up for a line-of-sight scenario

## **Appendix IV**

### **RF EMF exposure limits**

(This appendix does not form an integral part of this Recommendation.)

Administrations are encouraged to follow the ICNIRP Guidelines [b-ICNIRP] or IEEE Standard [b-IEEE C95.1], or science-based RF EMF exposure limits set by their own experts. The best practice for Administrations that choose to use international RF EMF exposure limits is to limit the exposure levels to the thresholds specified in [b-ICNIRP]. [b-ICNIRP] and [b-IEEE C95.1] limits are largely harmonized, and the power-density limits for whole-body exposure to continuous fields are identical above 30 MHz. See [ITU-T K.91].

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